Newtonian Gravity with Holonomic correction

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EQFT Unified Fields – Newtonian Gravity with Holonomic correction

Abstract

Everonic Quantum Field Theory (EQFT) represents a generalized, deterministic quantum gravitational framework that unifies holonomic gauge fields, spacetime curvature, and photonic quantum information systems. It synthesizes topological field theory, nonlinear optics, and quantum communication protocols into a single cohesive model. EQFT is built upon the principles of 1) Collapse-free quantum entanglement and holonomic evolution, 2) Gravitational field coupling via modified Einstein equations, 3) Non-abelian gauge dynamics and self-duality, 4) Quantum optical phenomena including SPDC (Spontaneous Parametric Down-Conversion), 5) Photonic manipulation through metasurfaces and waveguides, 6) Real-time feedback control and error-correction using stabilizer codes, and 7) Scalability via modular SU(n)-based quantum gates and routing algorithms. This framework extends and elevates both Einstein spacetime theory and the Seiberg–Witten topological gauge field model, giving rise to a reproducible, room-temperature, holographically expandable unified field theory. **Newtonian gravity** with the **Everon Quantum Field Theory (EQFT)** framework, recognizes that both describe gravity — from **complementary perspectives**.

Keywords: Holonomy, Newtonian Dynamics, Levitation, Localized holonomy, Unified Field Theory, Everon Quantum Field Theory, geometric gravity, Holonomic curvature.

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Introduction

Everonic Unified fields

Everon Lagrangian (Full Field Inclusion)

$$\mathscr{L}_{\text{EQFT}} = \mathscr{L}_{\text{grav}} + \mathscr{L}_{\text{spinor}} + \mathscr{L}_{\text{holo}} + \mathscr{L}_{\text{photon}} + \mathscr{L}_{\text{phonon}} + \mathscr{L}_{\text{int}}$$

Encapsulates all quantum, holonomic, acoustic, gravitational, and interaction terms within the Everon field manifold.

Everon Hamiltonian (Full Field Inclusion)

$$\mathscr{H}_{\text{EQFT}} = \mathscr{H}_{\text{grav}} + \mathscr{H}_{\text{spinor}} + \mathscr{H}_{\text{holo}} + \mathscr{H}_{\text{photon}} + \mathscr{H}_{\text{phonon}} + \mathscr{H}_{\text{int}}$$

The total Hamiltonian corresponding to the Lagrangian formulation, expressed via Legendre transformation and supporting holonomic dynamics. Hamiltonian, Lagrangian and Newtonian solutions emerge from the unification of the following theory and quantum field frameworks; we begin with:

1. Seiberg–Witten Equations (Topological Gauge Theory): $D_A\psi = 0 \qquad F_A^+ = \sigma(\psi,\psi)$.

2. Holonomic Gauge Fields: $\mathscr{A}_\mu = \langle\psi\,|\,\mathscr{U}_\gamma^\dagger \nabla_\mu \mathscr{U}_\gamma\,|\,\psi\rangle, \quad F_{\mu\nu}^{(\mathscr{A})} = \partial_\mu \mathscr{A}_\nu - \partial_\nu \mathscr{A}_\mu + [\mathscr{A}_\mu, \mathscr{A}_\nu]$

3. Bogoliubov Transformation (Photonic/Quantum Optics): $a_k' = u_k a_k + v_k b_k^\dagger$

4. Phonon Field Theory (Acoustic Quantum Fields):

$$\mathscr{L}_{\text{phonon}} = \frac{1}{2}\left(\partial^\mu \phi_{\text{ac}} \partial_\mu \phi_{\text{ac}} - V(\phi_{\text{ac}})\right), \quad V(\phi_{\text{ac}}) = \frac{1}{2} m^2 \phi_{\text{ac}}^2 + \lambda \phi_{\text{ac}}^4$$

5. Gravitational Fields (Einstein–Cartan Formalism): $\mathscr{L}_{\text{grav}} = \frac{1}{2\kappa}\sqrt{-g}(R + \Lambda)$

6. Quantum Holonomic Phase Loop Equation: $\phi(t) = \oint_C \mathscr{A}_\mu dx^\mu = \gamma_{\text{hol}}(t)$

7. Newtonian Limit with Holonomic Correction (Acoustic and Navigation-Enabled):

$$m\ddot{x} = -\nabla V_{\text{eff}} + \hbar \frac{d}{dt}(\nabla \gamma_{\text{hol}})$$

8. Transparency Equation (Quantum Field Opacity): $O(x,t) = 1 - \alpha \left| E(x,t) e^{i\gamma \mathrm{hol}^{(x,t)}} \right|^2$

9. Biofield Extraction and Filtering: $|\Phi(t)\rangle = \sum_i c_i(t) |f_i\rangle \quad \rightarrow \quad |\Phi'(t)\rangle = \sum_{i \in H} c_i(t) |f_i\rangle$

10. Quantum Power Extraction: $P(t) = \sum_n \eta_n \left| E_n(t) e^{i\gamma_n(t)} \right|^2$

11. Elemental Extraction Phase Encoding: $E_{\mathrm{extract}}(x,t) = E_0 \cdot e^{i\gamma_k(x,t)}$

where, $\gamma_k(x,t) = \arg(\chi_k(x,t))$

12. Alcubierre Warp Metric: $ds^2 = -c^2 dt^2 + [dx - v_s(t) f(r_s) dt]^2 + dy^2 + dz^2$

13. Geodesic Equation (Spacetime Trajectories): $\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0$

14. Bogoliubov Metasurface Transformations:

$$\hat{A}_j = \sum_k \left(U_{jk} \hat{a}_k + V_{jk} \hat{a}_k^\dagger \right)$$

$$H = \sum_k \epsilon_k \hat{a}_k^\dagger \hat{a}_k + \sum_k \Delta_k \hat{a}_k^\dagger \hat{a}_{-k}^\dagger + \mathrm{h.c.}$$

$$\Psi_{\mathrm{out}}(x,t) = \mathscr{M}(x,t) \cdot \left(u_k \Psi_{\mathrm{in}} + v_k \Psi_{\mathrm{in}}^* \right)$$

By unifying these sets of equations, we can freely navigate the frameworks.

Unified Lagrangian and Newtonian Formulation

The following section details the integration of classical Newtonian mechanics into the Everon Quantum Field Theory (EQFT) framework, with special attention to Newtonian Dynamics, holonomic phase corrections, gravitational definitions, and compatibility within quantum fields.

Everonic Unified Lagrangian (Full Field Inclusion)

$$L_{EQFT} = L_{grav} + L_{spinor} + L_{holo} + L_{photon} + L_{phonon} + L_{int}$$

Definitions:

L_{grav}: Gravitational Lagrangian (Einstein–Cartan curvature term) [1]

L_{spinor}: Fermionic field term with spin connection [2]

L_{holo}: Holonomic gauge contribution from SU(2)/SU(3) loops [3]

L_{photon}: Electromagnetic field tensor term [4]

L_{phonon}: Acoustic quantum field term [5]

L_{int}: Interaction terms between fields and metasurfaces [6]

Everonic Hamiltonian (Canonical Form)

$$H_{EQFT} = H_{grav} + H_{spinor} + H_{holo} + H_{photon} + H_{phonon} + H_{int}$$

Each component mirrors its Lagrangian counterpart but is expressed in canonical momenta [7].

Everonic Newtonian Gravity Equation Reconciliation

$$m\ddot{x} = -\nabla V_{\text{eff}} + \hbar \frac{d}{dt}(\nabla \gamma_{\text{hol}})$$

Definitions:

m: Particle or object mass

$\ddot{x}$: Acceleration (second derivative of position with respect to time)

$V_{\text{eff}} = V_g + V_{\text{meta}}$

V_g: Newtonian gravitational potential (−GM/r)

G: Newtonian gravitational constant

M: Mass of gravitational source

r: Distance to mass center

V_{meta}: Holonomic/metasurface potential [9]

γ_{hol}: Holonomic phase accumulated over SU(2)/SU(3) loop [10]

ħ: Reduced Planck's constant (ħ = h / 2π)

Gravitational Force Formulation

$$\vec{F}_g = -G\frac{m_1 m_2}{r^2} \quad \Rightarrow \quad m\ddot{x} = -\nabla V_g$$

Definitions:

$\vec{F}_g$: Classical gravitational force vector

G: Newtonian gravitational constant

m_1, m_2: Interacting masses

r : Distance between mass centers

$\ddot{x}$: Acceleration of mass m in field V_g

Compatibility Note

In EQFT, Newtonian gravity is fully preserved as a special case. Holonomic correction introduces a local geometric field effect similar to inertial force, while preserving locality and deterministic motion.

Quantitative Behavior

Case 1 ($\gamma_{hol} \to 0$):

Correction vanishes. Equation reduces to: $m\ddot{x} = -\nabla V_g$

Case 2 ($\gamma_{hol} \to \gamma_{classical}$):

Holonomic phase approximates classical behavior. System exhibits metasurface-induced acceleration without violating conservation laws.

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